

To setting up minikube on ubuntu 24.04 and exploring Kubernetes namespaces

Prerequisites:

Requirement	Notes
OS	Ubuntu 24.04 LTS
CPU	2 vCPUs minimum
RAM	4 GB minimum (8 GB recommended)
Disk	20 GB free space
Container/VM runtime	Docker (recommended driver)
User privileges	sudo access

Step 01: Update the Ubuntu 24.04 LTS system after deploying a new VM and installing OS version.

```
student@minikubesrv: ~  
login as: student  
student@192.168.110.130's password:  
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-31-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
This system has been minimized by removing packages and content that are  
not required on a system that users do not log into.  
  
To restore this content, you can run the 'unminimize' command.  
student@minikubesrv:~$
```

\$ sudo apt update && sudo apt upgrade -y && sudo reboot

```
student@minikubesrv:~$ sudo apt update && sudo apt upgrade -y && sudo reboot  
[sudo] password for student:  
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:2 http://in.archive.ubuntu.com/ubuntu noble InRelease [256 kB]  
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827  
kB]  
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:5 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:6 http://in.archive.ubuntu.com/ubuntu noble/main amd64 Packages [1401 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [187  
kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [44  
.8 kB]  
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages  
[1133 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-e  
n [259 kB]  
Get:11 http://in.archive.ubuntu.com/ubuntu noble/main Translation-en [513 kB]  
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Compone  
nts [212 B]  
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages  
[1174 kB]  
Get:14 http://in.archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [93.9  
kB]  
Get:15 http://in.archive.ubuntu.com/ubuntu noble/restricted Translation-en [18.7  
kB]  
Get:16 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 M  
B]  
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en  
[231 kB]
```

Step 02: Install Docker (container runtime / driver) after adding the docker repository and docker's official GPG key,

```
$ echo \
```

```
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
```

```
https://download.docker.com/linux/ubuntu \
```

```
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
```

```
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
student@minikubesrv:~$ echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
[sudo] password for student:
student@minikubesrv:~$ sudo apt install -y ca-certificates curl gnupg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~24.04.1).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.11).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
The following packages were automatically installed and are no longer required:
  libtirpc-common libtirpc3t64 python3-netifaces
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
$ sudo apt install -y ca-certificates curl gnupg
```

```
$ sudo apt update; sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
```

```
sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [60.5 kB]
Hit:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 109 kB in 2s (62.5 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-ce is already the newest version (5:29.6.1~1-ubuntu.24.04~noble).
docker-ce-cli is already the newest version (5:29.6.1~1-ubuntu.24.04~noble).
containerd.io is already the newest version (2.2.6~1-ubuntu.24.04~noble).
docker-buildx-plugin is already the newest version (0.35.0~1-ubuntu.24.04~noble).
docker-compose-plugin is already the newest version (5.3.1~1-ubuntu.24.04~noble).
The following packages were automatically installed and are no longer required:
  libtirpc-common libtirpc3t64 python3-netifaces
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Step 03: Allow your user to run docker without sudo and verify:

```
$ sudo docker version
```

```
student@minikubesrv:~$ docker version
Client: Docker Engine - Community
 Version:           29.6.1
 API version:       1.55
 Go version:        gol.26.4
 Git commit:        8900f1d
 Built:             Fri Jun 26 11:40:19 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:           29.6.1
  API version:       1.55 (minimum version 1.40)
  Go version:        gol.26.4
  Git commit:        89c5ab3
  Built:             Fri Jun 26 11:40:19 2026
  OS/Arch:           linux/amd64
  Experimental:      false
 containerd:
  Version:           v2.2.6
  GitCommit:        11ce9d5f3c68c941867e82890e93e815c1304f1b
 runc:
  Version:           1.3.6
  GitCommit:        v1.3.6-0-g491b69ba
 docker-init:
  Version:           0.19.0
  GitCommit:        de40ad0
```

```
student@minikubesrv:~$ sudo usermod -aG docker $USER; newgrp docker
```


Step 05: Access the minikube dashboard:

\$ minikube dashboard

```
student@minikubesrv:~$ kubectl get pods -A
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE
kube-system   coredns-7d764666f9-48rjb             1/1     Running   0           3m27s
kube-system   etcd-minikube                         1/1     Running   0           3m37s
kube-system   kube-apiserver-minikube               1/1     Running   0           3m32s
kube-system   kube-controller-manager-minikube     1/1     Running   0           3m32s
kube-system   kube-proxy-8fqjb                     1/1     Running   0           3m27s
kube-system   kube-scheduler-minikube              1/1     Running   0           3m35s
kube-system   storage-provisioner                  1/1     Running   1 (2m59s ago) 3m25s
student@minikubesrv:~$ minikube dashboard
* Enabling dashboard ...
  - Using image docker.io/kubernetesui/dashboard:v2.7.0
  - Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
* Some dashboard features require the metrics-server addon. To enable all features please run:
    minikube addons enable metrics-server
* Verifying dashboard health ...
* Launching proxy ...
* Verifying proxy health ...
* Opening http://127.0.0.1:38129/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
http://127.0.0.1:38129/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/
```

Step 06: Work with Namespaces (list namespaces, create a namespace and use namespace in deployments):

\$ kubectl get namespaces

```
student@minikubesrv:~$ kubectl get namespaces
NAME                                STATUS    AGE
default                            Active    12m
dev                                Active    110s
kube-node-lease                    Active    12m
kube-public                        Active    12m
kube-system                        Active    12m
kubernetes-dashboard               Active    7m50s
student@minikubesrv:~$
```

\$ kubectl create namespace dev

```
No resources found in development namespace.
student@minikubesrv:~$ kubectl get all -n dev
NAME                                READY   STATUS    RESTARTS   AGE
pod/hello-minikube-7b759d4c76-p4n9q 0/1     ImagePullBackOff 0           30m
pod/nginx-56c45fd5ff-d8vfl          1/1     Running    1 (13m ago) 71m

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/hello-minikube              NodePort      10.108.93.72  <none>         8080:30007/TCP   35m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/hello-minikube      0/1     1             0           30m
deployment.apps/nginx               1/1     1             1           71m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/hello-minikube-7b759d4c76 1         1         0       30m
replicaset.apps/nginx-56c45fd5ff          1         1         1       71m
```

\$ kubectl get svc -n dev

\$ kubectl create deployment nginx --image=nginx -n dev

\$ minikube service hello-minikube -n dev

```
student@minikubesrv:~$ kubectl get svc -n dev
NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
hello-minikube                     NodePort      10.108.93.72  <none>         8080:30007/TCP   4m47s
student@minikubesrv:~$ minikube service hello-minikube -n dev
```

NAMESPACE	NAME	TARGET PORT	URL
dev	hello-minikube	8080	http://192.168.49.2:30007

X Exiting due to SVC_UNREACHABLE: service not available: no running pod for service hello-minikube found

```
* If the above advice does not help, please let us know:
  https://github.com/kubernetes/minikube/issues/new/choose

* Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.
* Please also attach the following file to the GitHub issue:
* - /tmp/minikube_service_1bc46e5603c963352b4f6be30e151df50339274b_0.log
```